MLOPS Assignment 2

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1. Which run had the lowest latency, and why might that be?

   run_batch8 had the lowest latency (0.120 s), followed by run_batch4 (0.133 s) and run_batch1 (0.226 s). With batch size 8, the model processes several sentences in one forward pass, so it makes fewer calls and has less overhead per call. run_batch1 makes 10 separate calls, and since it ran first it also paid a one-time warm-up cost

2. Did accuracy change between runs? Why or why not?

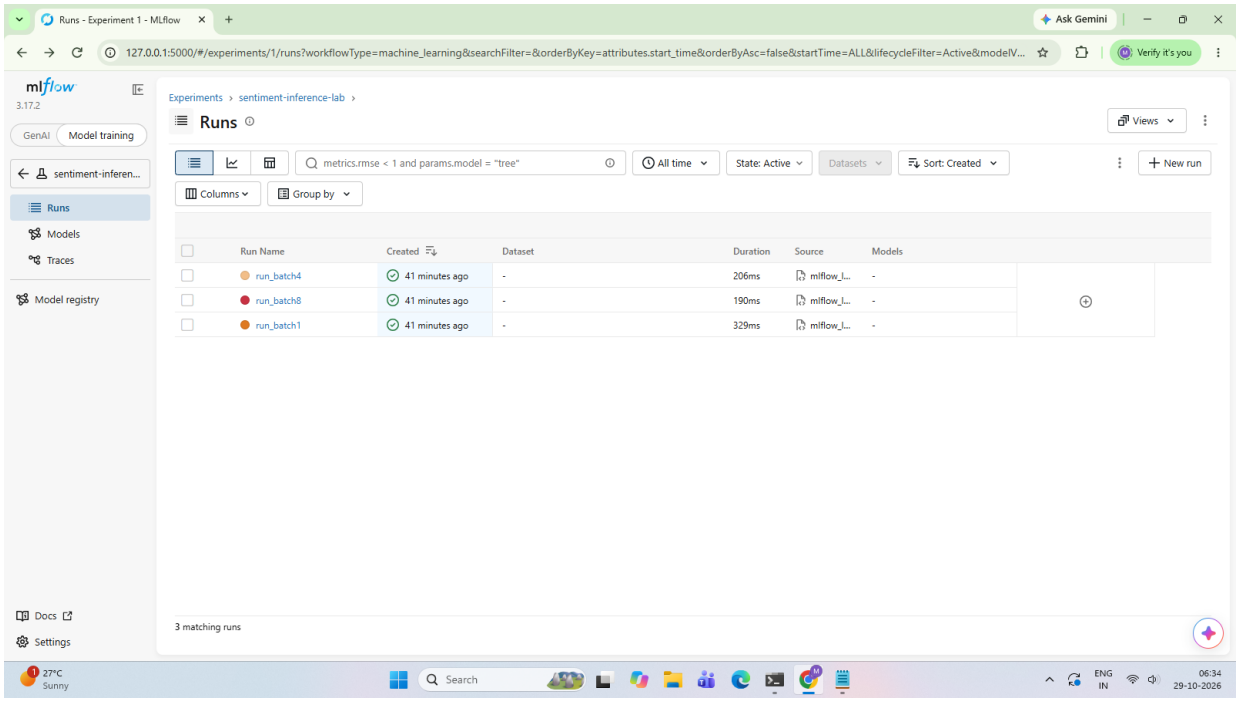
   No, accuracy stayed at 1.0 (100%) in all three runs. Every run used the same pre-trained model, with the same weights, on the same 10 sentences, and inference is deterministic. Batch size only changes how fast the sentences are processed, not the predictions

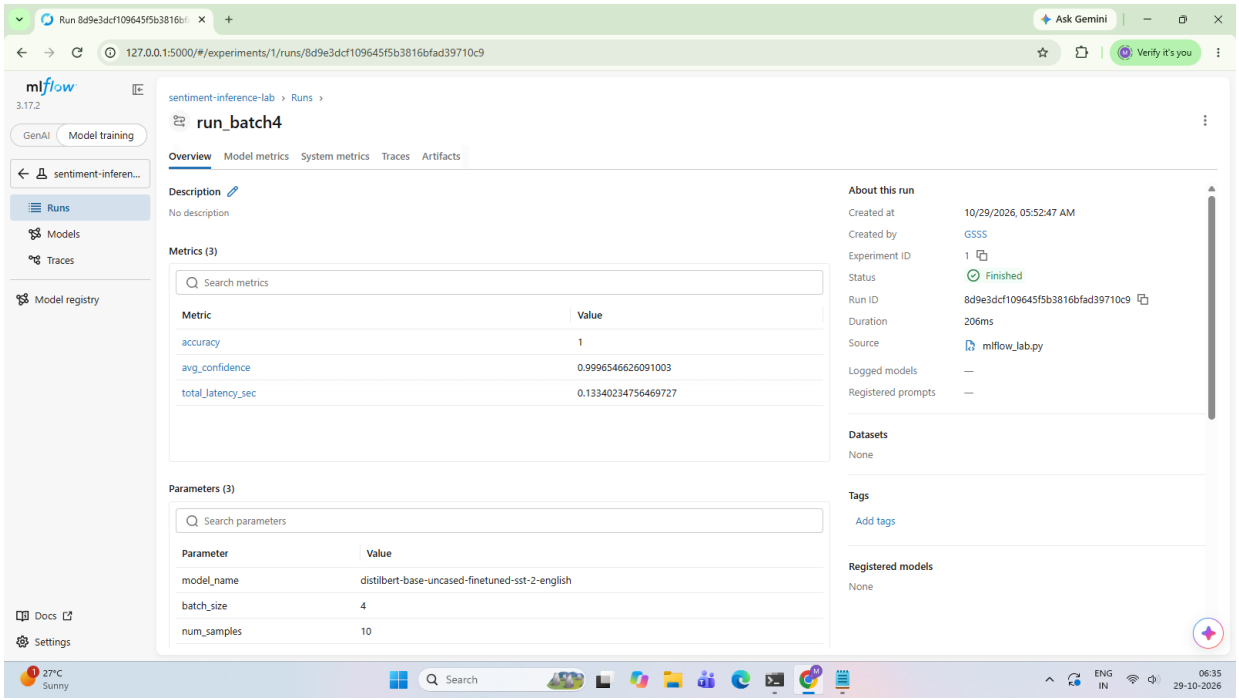
3. Name two benefits of using MLflow here instead of printing results in the terminal.

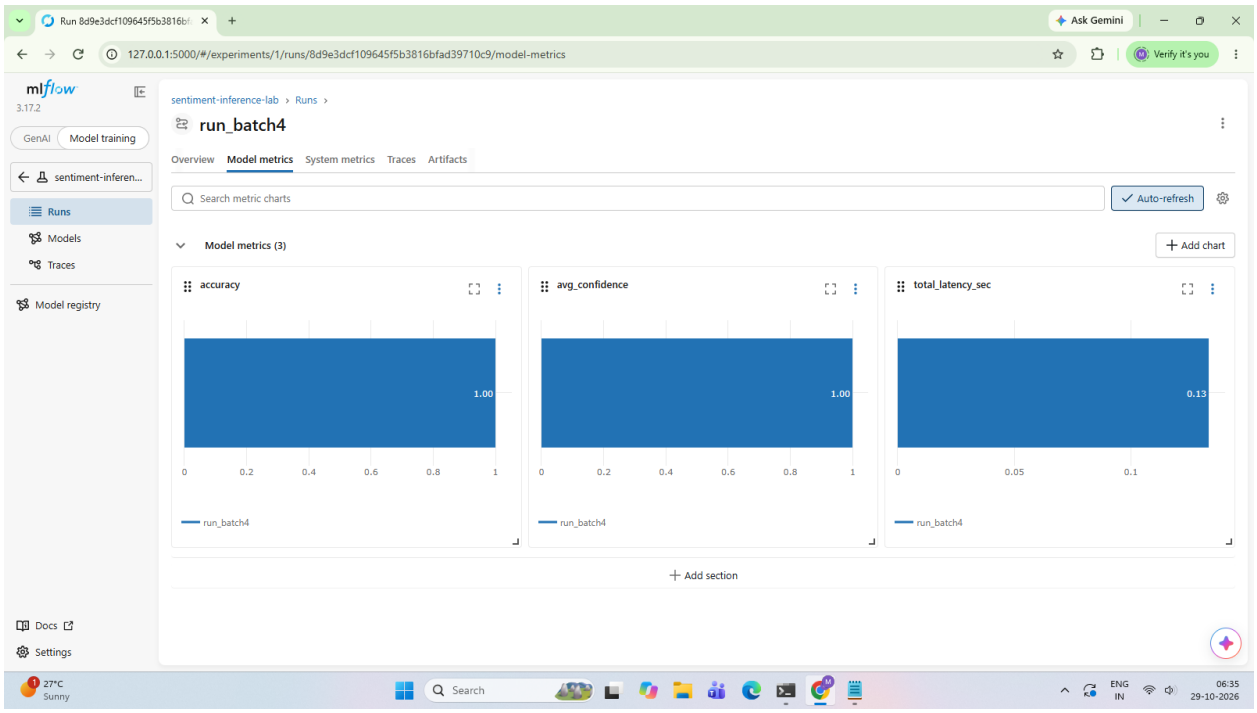
   Two benefits of MLflow:

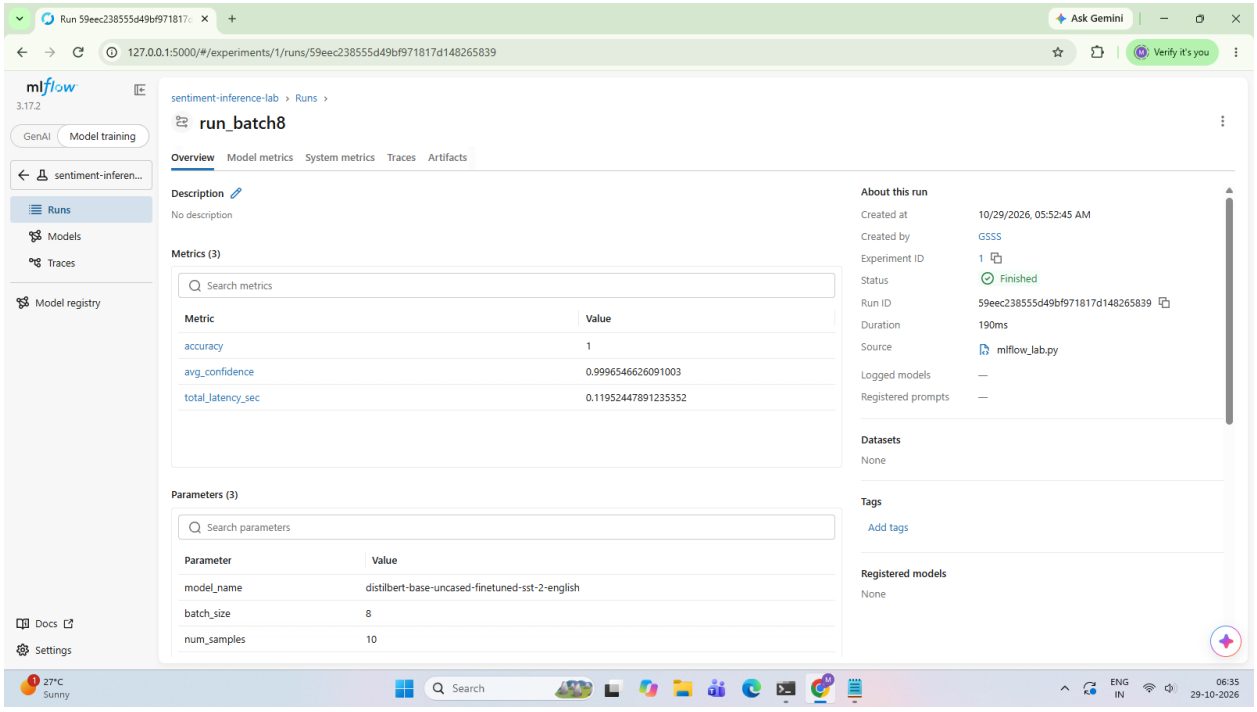
   - Reproducibility and record-keeping: the parameters, metrics and artifacts (predictions.csv) of every run are saved permanently. Terminal output is lost once the window closes.
   - Easy comparison: you can view and compare runs side by side in the UI, with tables and charts, instead of reading through printed lines

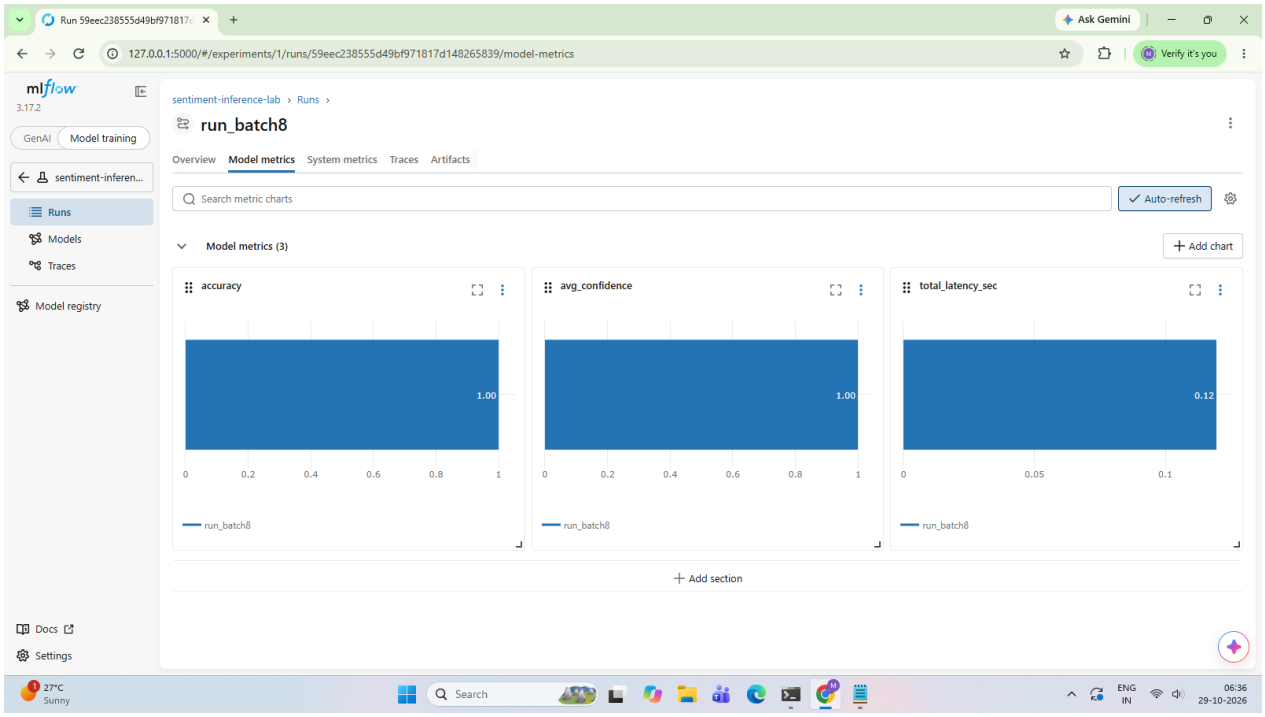
Screenshot of the MLflow runs table:

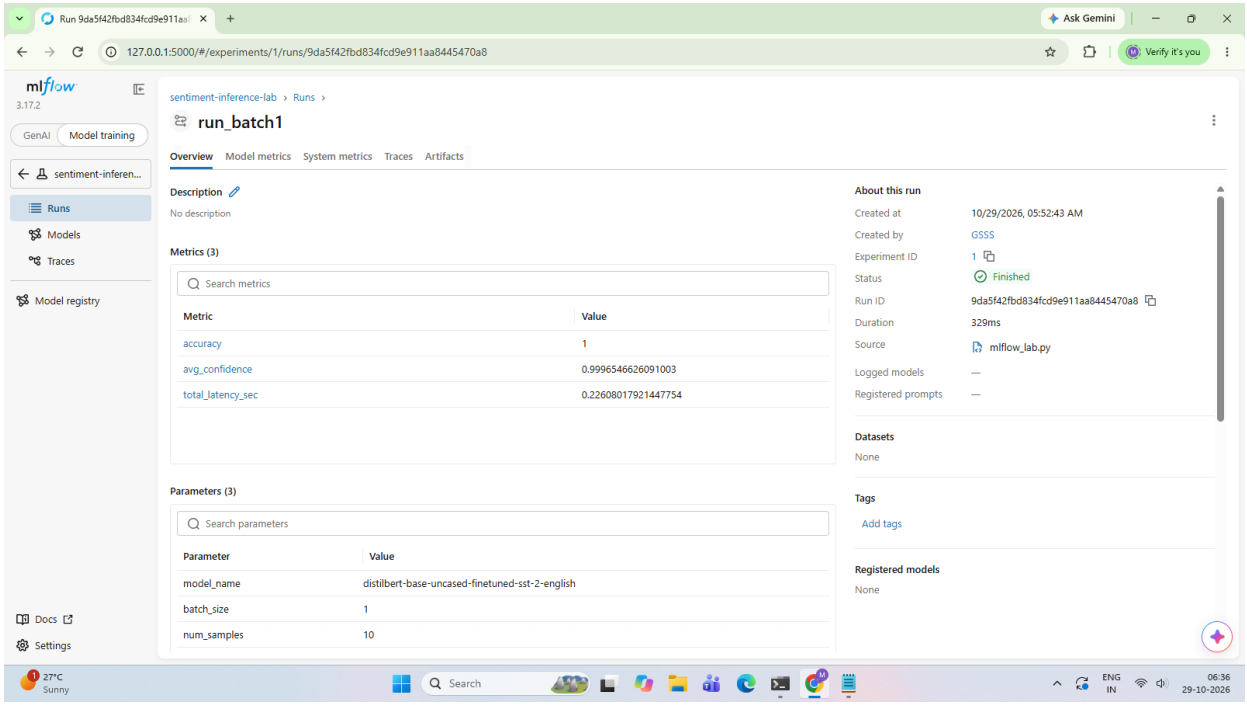


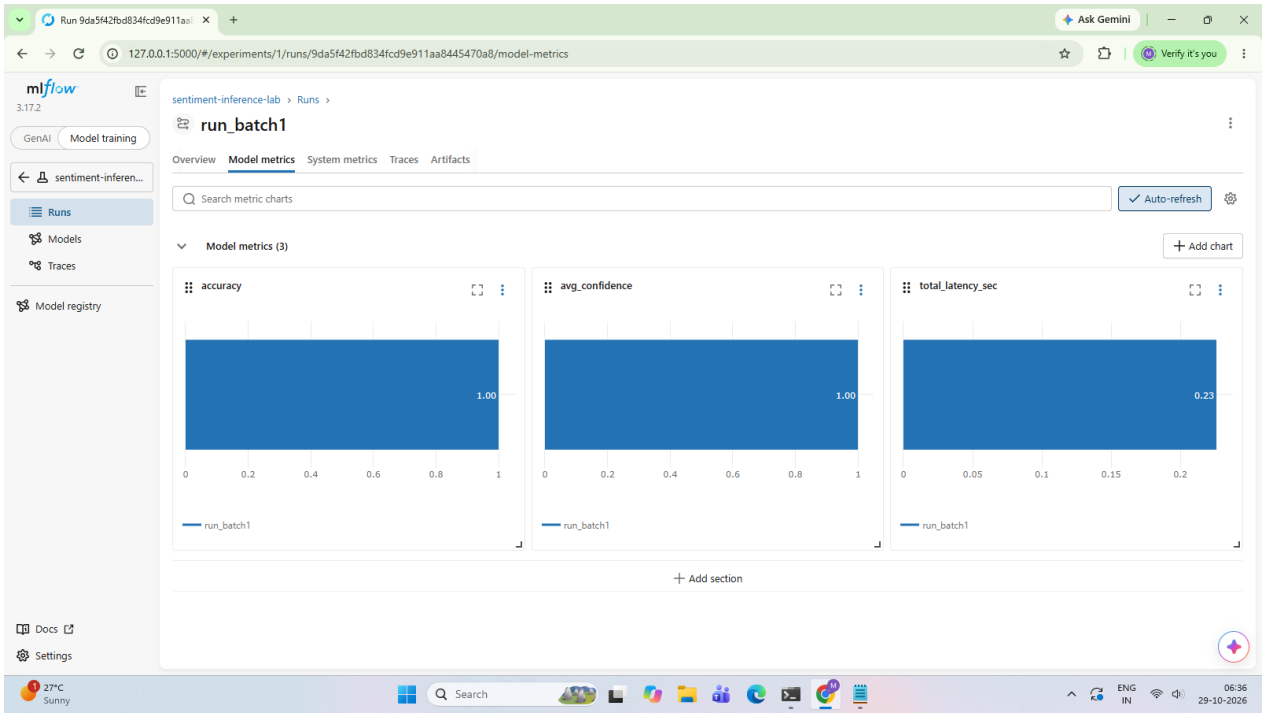



Run 8d9e3dcf109645f5b3816bf
Ask Gemini
127.0.0.1:5000/#/experiments/1/runs/8d9e3dcf109645f5b3816bfad39710c9/model-metrics
Verify it's you
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch4
Overview
Model metrics
System metrics
Traces
Artifacts
Search metric charts
Auto-refresh
Model metrics (3)
Add chart
accuracy
avg_confidence
total_latency_sec
1.00
1.00
0.13
0
0.2
0.4
0.6
0.8
1
0.05
0.1
run_batch4
Add section
27°C
Sunny
Search
ENG
IN
06:35
29-10-2026

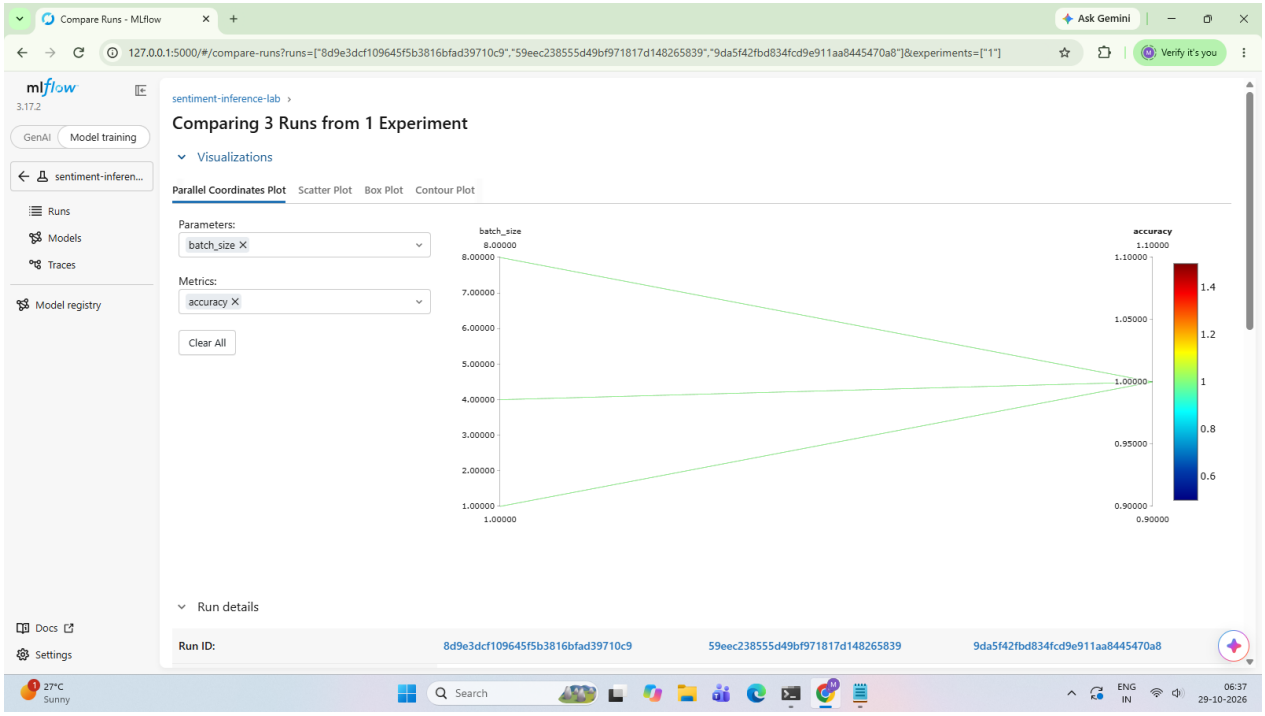

Run 59eec238555d49bf971817c
Ask Gemini
127.0.0.1:5000/#/experiments/1/runs/59eec238555d49bf971817d148265839
Verify it's you
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
Description
No description
Metrics (3)
Search metrics
Metric
Value
accuracy
1
avg_confidence
0.9996546626091003
total_latency_sec
0.11952447891235352
Parameters (3)
Search parameters
Parameter
Value
model_name
distilbert-base-uncased-finetuned-sst-2-english
batch_size
8
num_samples
10
About this run
Created at
10/29/2026, 05:52:45 AM
Created by
GSSS
Experiment ID
1
Status
Finished
Run ID
59eec238555d49bf971817d148265839
Duration
190ms
Source
mlflow_lab.py
Logged models
—
Registered prompts
—
Datasets
None
Tags
Add tags
Registered models
None
27°C
Sunny
Search
ENG
IN
06:35
29-10-2026


Run 59eec238555d49bf971817
Ask Gemini
127.0.0.1:5000/#/experiments/1/runs/59eec238555d49bf971817d148265839/model-metrics
Verify it's you
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
Search metric charts
Auto-refresh
Model metrics (3)
Add chart
accuracy
avg_confidence
total_latency_sec
1.00
1.00
0.12
0
0.2
0.4
0.6
0.8
1
0.05
0.1
run_batch8
Add section
27°C
Sunny
Search
ENG
IN
06:36
29-10-2026

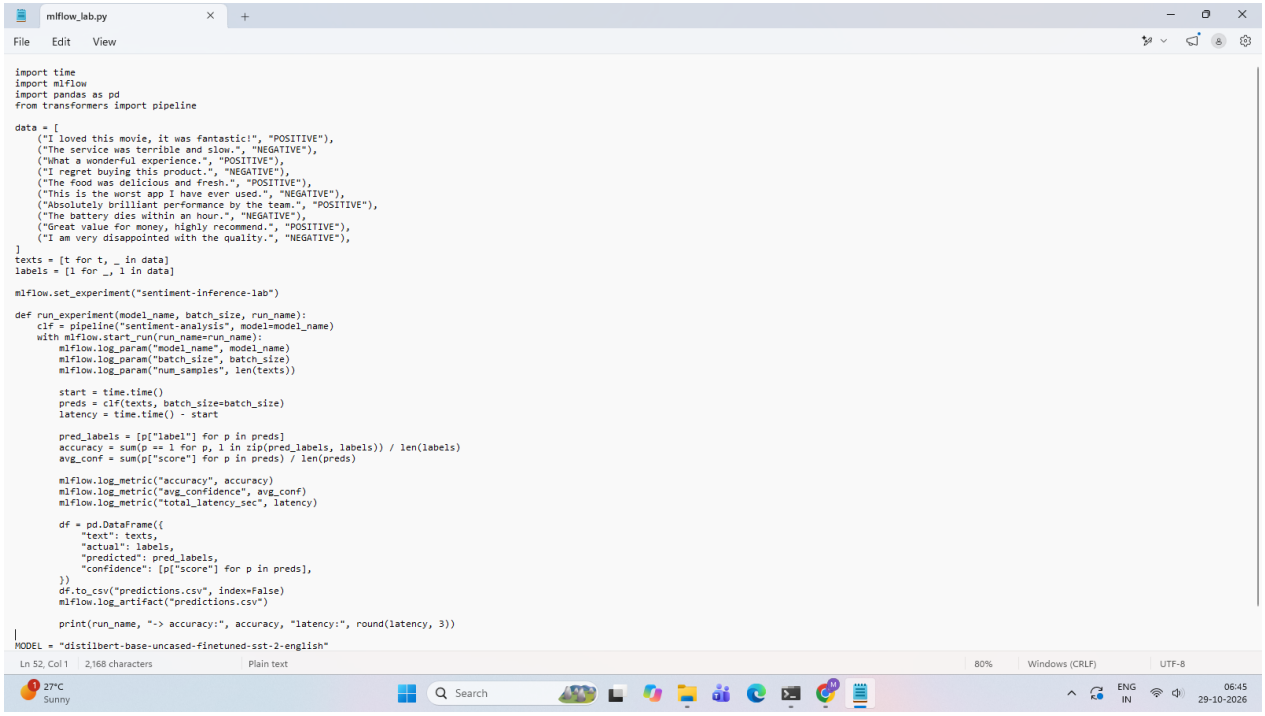

Run 9da5f42fbd834fcd9e911aa
Ask Gemini
127.0.0.1:5000/#/experiments/1/runs/9da5f42fbd834fcd9e911aa8445470a8
Verify it's you
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch1
Overview
Model metrics
System metrics
Traces
Artifacts
Description
No description
Metrics (3)
Search metrics
Metric
Value
accuracy
1
avg_confidence
0.9996546626091003
total_latency_sec
0.22608017921447754
Parameters (3)
Search parameters
Parameter
Value
model_name
distilbert-base-uncased-finetuned-sst-2-english
batch_size
1
num_samples
10
About this run
Created at
10/29/2026, 05:52:43 AM
Created by
GSSS
Experiment ID
1
Status
Finished
Run ID
9da5f42fbd834fcd9e911aa8445470a8
Duration
329ms
Source
mlflow_lab.py
Logged models
—
Registered prompts
—
Datasets
None
Tags
Add tags
Registered models
None
27°C
Sunny
Search
ENG
IN
06:36
29-10-2026



Screenshot of the Compare view:



mlflow_lab.py file:



```
import time
import mlflow
import pandas as pd
from transformers import pipeline

data = [
    ("I loved this movie, it was fantastic!", "POSITIVE"),
    ("The service was terrible and slow.", "NEGATIVE"),
    ("What a wonderful experience.", "POSITIVE"),
    ("I regret buying this product.", "NEGATIVE"),
    ("The food was delicious and fresh.", "POSITIVE"),
    ("This is the worst app I have ever used.", "NEGATIVE"),
    ("Absolutely brilliant performance by the team.", "POSITIVE"),
    ("The battery dies within an hour.", "NEGATIVE"),
    ("Great value for money, highly recommend.", "POSITIVE"),
    ("I am very disappointed with the quality.", "NEGATIVE"),
]
texts = [t for t, _ in data]
labels = [1 for _, l in data]

mlflow.set_experiment("sentiment-inference-lab")

def run_experiment(model_name, batch_size, run_name):
    clf = pipeline("sentiment-analysis", model=model_name)
    with mlflow.start_run(run_name=run_name):
        mlflow.log_param("model_name", model_name)
        mlflow.log_param("batch_size", batch_size)
        mlflow.log_param("num_samples", len(texts))

        start = time.time()
        preds = clf(texts, batch_size=batch_size)
        latency = time.time() - start

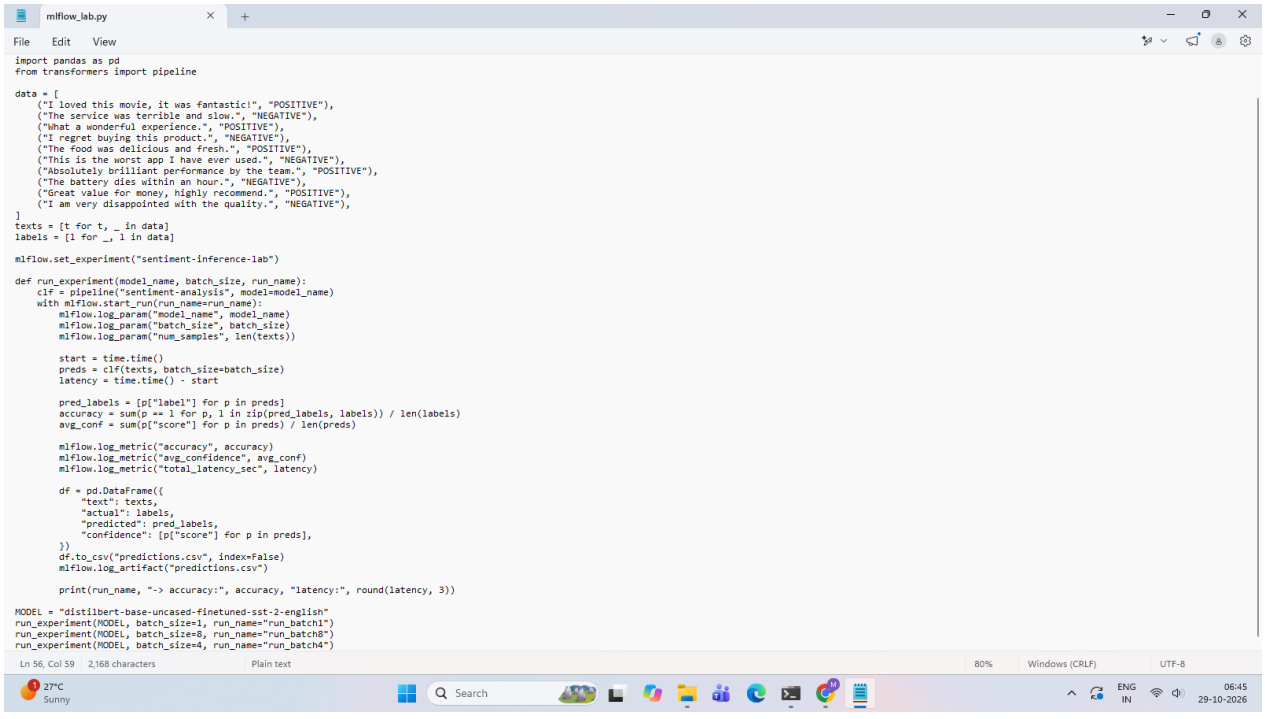
        pred_labels = [p["label"] for p in preds]
        accuracy = sum(p == l for p, l in zip(pred_labels, labels)) / len(labels)
        avg_conf = sum(p["score"] for p in preds) / len(preds)

        mlflow.log_metric("accuracy", accuracy)
        mlflow.log_metric("avg_confidence", avg_conf)
        mlflow.log_metric("total_latency_sec", latency)

        df = pd.DataFrame({
            "text": texts,
            "actual": labels,
            "predicted": pred_labels,
            "confidence": [p["score"] for p in preds],
        })
        df.to_csv("predictions.csv", index=False)
        mlflow.log_artifact("predictions.csv")

        print(run_name, "-> accuracy:", accuracy, "latency:", round(latency, 3))

MODEL = "distilbert-base-uncased-finetuned-sst-2-english"
```



```
import pandas as pd
from transformers import pipeline

data = [
    ("I loved this movie, it was fantastic!", "POSITIVE"),
    ("The service was terrible and slow.", "NEGATIVE"),
    ("What a wonderful experience.", "POSITIVE"),
    ("I regret buying this product.", "NEGATIVE"),
    ("The food was delicious and fresh.", "POSITIVE"),
    ("This is the worst app I have ever used.", "NEGATIVE"),
    ("Absolutely brilliant performance by the team.", "POSITIVE"),
    ("The battery dies within an hour.", "NEGATIVE"),
    ("Great value for money, highly recommend.", "POSITIVE"),
    ("I am very disappointed with the quality.", "NEGATIVE"),
]
texts = [t for t, _ in data]
labels = [1 for _, l in data]

mlflow.set_experiment("sentiment-inference-lab")

def run_experiment(model_name, batch_size, run_name):
    clf = pipeline("sentiment-analysis", model=model_name)
    with mlflow.start_run(run_name=run_name):
        mlflow.log_param("model_name", model_name)
        mlflow.log_param("batch_size", batch_size)
        mlflow.log_param("num_samples", len(texts))

        start = time.time()
        preds = clf(texts, batch_size=batch_size)
        latency = time.time() - start

        pred_labels = [p["label"] for p in preds]
        accuracy = sum(p == l for p, l in zip(pred_labels, labels)) / len(labels)
        avg_conf = sum(p["score"] for p in preds) / len(preds)

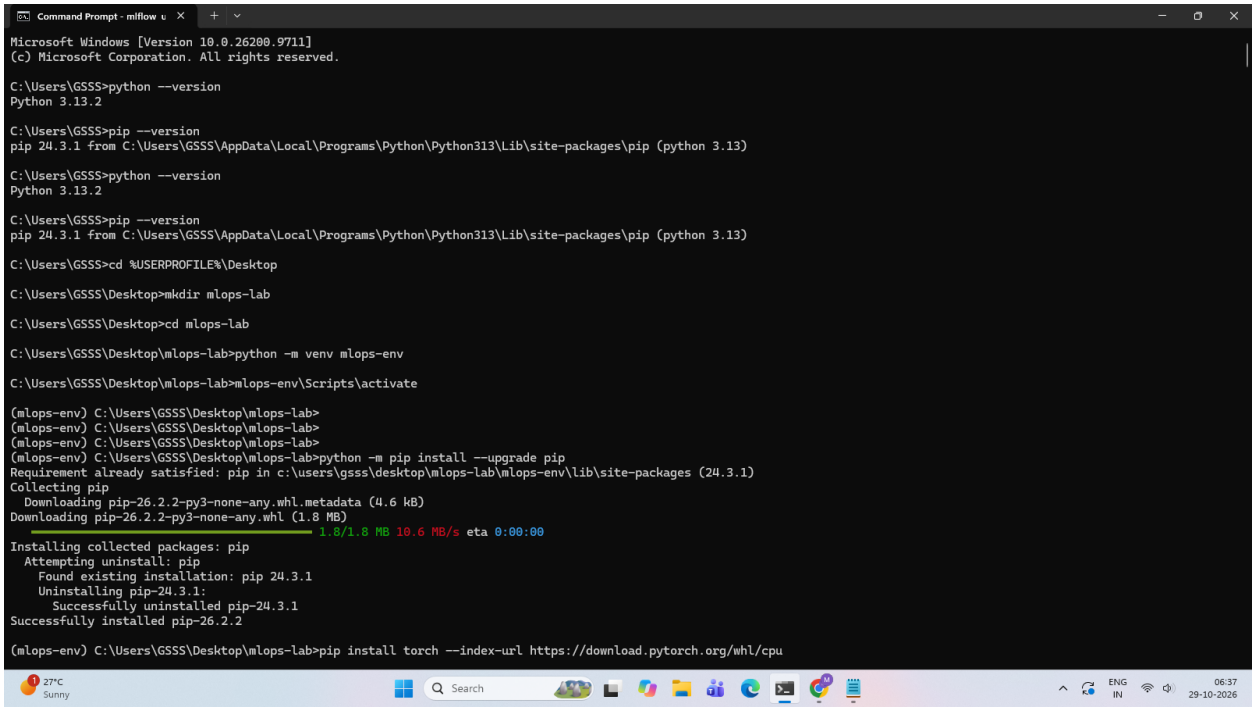
        mlflow.log_metric("accuracy", accuracy)
        mlflow.log_metric("avg_confidence", avg_conf)
        mlflow.log_metric("total_latency_sec", latency)

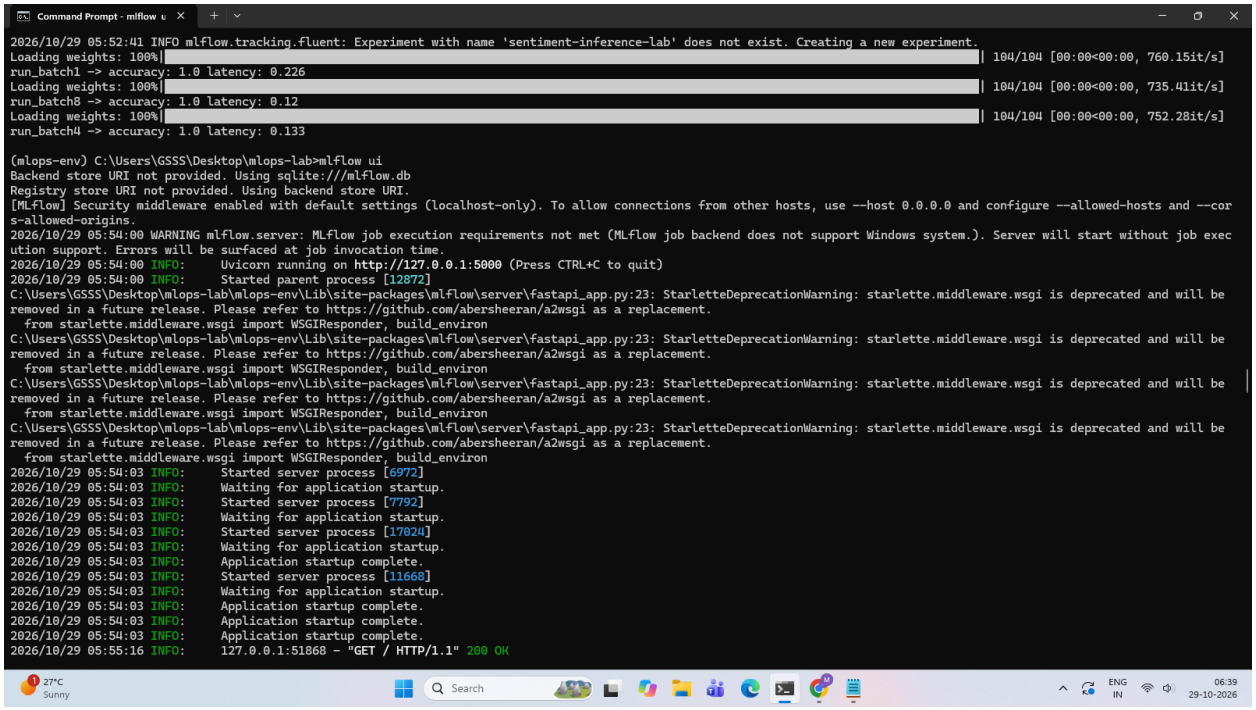
        df = pd.DataFrame({
            "text": texts,
            "actual": labels,
            "predicted": pred_labels,
            "confidence": [p["score"] for p in preds],
        })
        df.to_csv("predictions.csv", index=False)
        mlflow.log_artifact("predictions.csv")

        print(run_name, "-> accuracy:", accuracy, "latency:", round(latency, 3))

MODEL = "distilbert-base-uncased-finetuned-sst-2-english"
run_experiment(MODEL, batch_size=1, run_name="run_batch1")
run_experiment(MODEL, batch_size=8, run_name="run_batch8")
run_experiment(MODEL, batch_size=4, run_name="run_batch4")
```

Execution:


Command Prompt - mlflow u
Microsoft Windows [Version 10.0.26200.9711]
(c) Microsoft Corporation. All rights reserved.
C:\Users\GSSS>python --version
Python 3.13.2
C:\Users\GSSS>pip --version
pip 24.3.1 from C:\Users\GSSS\AppData\Local\Programs\Python\Python313\Lib\site-packages\pip (python 3.13)
C:\Users\GSSS>python --version
Python 3.13.2
C:\Users\GSSS>pip --version
pip 24.3.1 from C:\Users\GSSS\AppData\Local\Programs\Python\Python313\Lib\site-packages\pip (python 3.13)
C:\Users\GSSS>cd %USERPROFILE%\Desktop
C:\Users\GSSS\Desktop>mkdir mlops-lab
C:\Users\GSSS\Desktop>cd mlops-lab
C:\Users\GSSS\Desktop\mlops-lab>python -m venv mlops-env
C:\Users\GSSS\Desktop\mlops-lab>mlops-env\Scripts\activate
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>python -m pip install --upgrade pip
Requirement already satisfied: pip in c:\users\gsss\desktop\mlops-lab\mlops-env\lib\site-packages (24.3.1)
Collecting pip
Downloading pip-26.2.2-py3-none-any.whl.metadata (4.6 kB)
Downloading pip-26.2.2-py3-none-any.whl (1.8 MB)
1.8/1.8 MB 10.6 MB/s eta 0:00:00
Installing collected packages: pip
Attempting uninstall: pip
Found existing installation: pip 24.3.1
Uninstalling pip-24.3.1:
Successfully uninstalled pip-24.3.1
Successfully installed pip-26.2.2
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>pip install torch --index-url https://download.pytorch.org/whl/cpu
27°C
Sunny
Search
ENG
IN
06:37
29-10-2026


Command Prompt - mlflow u
2026/10/29 05:52:41 INFO mlflow.tracking.fluent: Experiment with name 'sentiment-inference-lab' does not exist. Creating a new experiment.
Loading weights: 100%| 104/104 [00:00<00:00, 760.15it/s]
run_batch1 -> accuracy: 1.0 latency: 0.226
Loading weights: 100%| 104/104 [00:00<00:00, 735.41it/s]
run_batch8 -> accuracy: 1.0 latency: 0.12
Loading weights: 100%| 104/104 [00:00<00:00, 752.28it/s]
run_batch4 -> accuracy: 1.0 latency: 0.133
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>mlflow ui
Backend store URI not provided. Using sqlite:///mlflow.db
Registry store URI not provided. Using backend store URI.
[MLflow] Security middleware enabled with default settings (localhost-only). To allow connections from other hosts, use --host 0.0.0.0 and configure --allowed-hosts and --cors-allowed-origins.
2026/10/29 05:54:00 WARNING mlflow.server: MLflow job execution requirements not met (MLflow job backend does not support Windows system.). Server will start without job execution support. Errors will be surfaced at job invocation time.
2026/10/29 05:54:00 INFO: Uvicorn running on http://127.0.0.1:5000 (Press CTRL+C to quit)
2026/10/29 05:54:00 INFO: Started parent process [12872]
C:\Users\GSSS\Desktop\mlops-lab\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
from starlette.middleware.wsgi import WSGIResponder, build_environ
2026/10/29 05:54:03 INFO: Started server process [6972]
2026/10/29 05:54:03 INFO: Waiting for application startup.
2026/10/29 05:54:03 INFO: Started server process [7792]
2026/10/29 05:54:03 INFO: Waiting for application startup.
2026/10/29 05:54:03 INFO: Started server process [17024]
2026/10/29 05:54:03 INFO: Waiting for application startup.
2026/10/29 05:54:03 INFO: Application startup complete.
2026/10/29 05:54:03 INFO: Started server process [11668]
2026/10/29 05:54:03 INFO: Waiting for application startup.
2026/10/29 05:54:03 INFO: Application startup complete.
2026/10/29 05:54:03 INFO: Application startup complete.
2026/10/29 05:54:03 INFO: Application startup complete.
2026/10/29 05:55:16 INFO: 127.0.0.1:51868 - "GET / HTTP/1.1" 200 OK
27°C
Sunny
Search
ENG
IN
06:39
29-10-2026